

		magnitude of the induced e.m.f. is larger when the magnet falls out of the solenoid.
26	D	Same heating effect implies same power dissipated in the resistor, i.e. $P = I^2 R$ For the new resistance $R_1 = R/4$, the new r.m.s. current I_1 is given by $P = I_1^2 R_1 = I_1^2 R/4 = I^2 R \rightarrow I_1^2 = 4I^2 \rightarrow I_1 = 2I$ The peak current $I_0 = \sqrt{2} I_1 = \sqrt{2} (2I)$
27	C	$E = [1.5 - (-2.5)] \times 1.6 \times 10^{-19} = 3.2 \times 10^{-19}$